

Ramnath Kumar

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RESEARCH INTERESTS & SUMMARY

My research centers on Deep Learning, with a particular emphasis on developing **robust** and **efficient** methods for building practical, real-world systems. I am especially interested in *representation learning* and in rethinking the optimization strategies that underpin modern training paradigms - seeking approaches that are not only scalable and generalizable but also adaptive to the diverse demands of deployment environments.

EDUCATION

University of California, Los Angeles (UCLA)

PhD. in Computer Science

. Advisor: [Prof. Cho-Jui Hsieh](#)

Los Angeles, USA

09/2024 – Present

BITS Pilani, Hyderabad Campus

B.E. in Computer Science, MSc. in Economics

. CGPA in Computer Science major: **9.65/10.0** (Top 10 in class of 255)

. Overall CGPA: 8.92/10.0

Hyderabad, India

08/2016 – 08/2021

SELECTED RESEARCH/WORK EXPERIENCE

Google Research

Student Researcher

Mountain View, USA

06/2026 – 09/2026

- . Mentors: [Dr. Fei Wang](#) and [Prof. Inderjit S. Dhillon](#)
- . Identified catastrophic forgetting in RL post-training on vision-language models, where out-of-domain capabilities collapse while suite-wide averages mask degradation. Developed Content-Free Anchoring (CFA), a Fisher-weighted MMD penalty using only synthetic anchors that reduces worst-decile forgetting to < 10% with finite-sample guarantees, outperforming curated-data and other traditional baselines [1].

Google DeepMind

Pre-Doctoral Researcher

Bengaluru, India

07/2022 – 08/2024

- . Mentors: [Dr. Prateek Jain](#) and [Prof. Inderjit S. Dhillon](#)
- . Led a cross-functional team in developing and deploying RGD in product-driven research, improving performance of ViT by up to **1.01%** on ImageNet-1K [5].
- . Developed an End-to-End efficient retrieval architecture (EHI), that improved upon prior retrieval benchmarks by up to **1.45%** at a fixed compute budget. This effort is slated for integration into various Google products [6].
- . Earned recognition in Google's AdsML organization for advancing scalable product retrieval models and optimization strategies, improving relevance and efficiency across global ad-serving systems through involving Related Ads features in the query of the model.
- . Took initiative in proposing a novel end-to-end retrieval architecture, improving efficiency and robustness in ongoing projects, with contributions recognized in . The approach has also been slated for integration with various teams including Keras, Youtube AutoML, and many more.

Google Research

Research Associate

Bengaluru, India

04/2022 – 07/2022

- . Mentor: [Dr. Dheeraj Nagaraj](#)
- . Developed Introspective Experience Replay (IER), a novel replay buffer sampling strategy inspired by Reverse Experience Replay (RER), demonstrating up to **7×** faster convergence in Reinforcement Learning algorithms such as DQN and TD3 [9].

Mila - Quebec Artificial Intelligence Institute

Consultant

Quebec, Canada

07/2021 – 03/2022

- . Mentor: [Prof. Yoshua Bengio](#)
- . Initiated and led a project exploring the impact of diversity in meta-learning, collaborating closely with his group. Designed and executed experiments, leading to an oral presentation at the “*Association for the Advancement of Artificial Intelligence*” (AAAI) that challenged the conventional wisdom of diversity being strictly useful to meta-learning [7].

Amazon ML

Applied Scientist Intern

Bengaluru, India

01/2021 – 06/2021

- . Mentor: [Dr. Gokul Swamy](#)
- . Led a study on causal attribution within Amazon’s sales model as first author, with findings presented at Amazon’s internal conference (AMLC 2021).

Kno.e.sis, Wright State University

Research Intern

Dayton, USA

05/2019 – 08/2019

- . Mentors: [Prof. Amit P. Sheth](#) and [Prof. Krishnaprasad Thirunarayan](#)
- . Collaborated with a cross-functional research team across multiple time zones as first author on the *eDarkFind* project, where I developed an unsupervised multi-view framework for Sybil detection, specifically targeting Darknet market activities [8].

PUBLICATIONS

* Indicates that the authors contributed equally to this work.

- [1] [Anchored in Noise: Mitigating Forgetting in RL Post-Training](#)
Ramnath Kumar, Jiong Zhang, Wei-Cheng Chang, Fei Wang, Hsiang-Fu Yu, Inderjit Dhillon, and Cho-Jui Hsieh
Under Review, ICLR 2027.
- [2] [FastLane: Efficient Routed Systems for Late-Interaction Retrieval](#)
Ramnath Kumar, Prateek Jain, and Cho-Jui Hsieh
Under Review, EACL, 2027.
- [3] [MIRAGE: Protecting against Malicious Image Editing via False Moderation](#)
Anshul Nasery, **Ramnath Kumar**, Cho-Jui Hsieh, and Sewoong Oh
Under Review, ICLR 2027.
- [4] [Flex-Act: Why Learn when you can Pick?](#)
Ramnath Kumar, Kyle Ritscher, Junmin Zhu, Lawrence Liu, and Cho-Jui Hsieh
Transactions on Machine Learning Research, 2026.
- [5] [Stochastic Re-weighted Gradient Descent via Distributionally Robust Optimization](#)
Ramnath Kumar, Kushal Alpesh Majmundar, Dheeraj Mysore Nagaraj, and Arun Suggala
Transactions on Machine Learning Research, 2024.
[Google AI Blog Coverage.](#)
- [6] [EHI: End-to-end Learning of Hierarchical Index for Efficient Dense Retrieval](#)
Ramnath Kumar*, Anshul Mittal*, Nilesh Gupta, Aditya Kusupati, Inderjit Dhillon, and Prateek Jain
Transactions on Machine Learning Research, 2024.
- [7] [The Effect of diversity in Meta-Learning](#) 
Ramnath Kumar, Tristan Deleu, and Yoshua Bengio
Association for the Advancement of Artificial Intelligence (AAAI), 2023 (Oral Paper).
[SyncedReview Blog Coverage.](#)
- [8] [eDarkFind: Unsupervised Multi-view Learning for Sybil Account Detection](#) 
Ramnath Kumar, Shweta Yadav, Raminta Daniulaityte, Francois Lamy, Krishnaprasad Thirunarayan, Usha Lokala, and Amit Sheth
The Web Conference (WWW), 2020.

- [9] [Introspective Experience Replay: Look Back When Surprised](#) 
Ramnath Kumar and Dheeraj Nagaraj
Transactions on Machine Learning Research, 2024.
[Google Research Blog Coverage](#).
- [10] [Rethinking Learning Dynamics in RL using Adversarial Networks](#) 
Ramnath Kumar, Tristan Deleu, and Yoshua Bengio
NeurIPS Workshop on DeepRL, 2022.
- [11] [Temporal Dynamics and Spatial Content in IoT Malware detection](#) 
Ramnath Kumar and G Geethakumari
IEEE TENCON, 2019.

SELECTED AWARDS AND HIGHLIGHTS

Quad Fellowship , Took part in a summit at the White House (U.S.A.) to discuss emerging technologies and their policy implications. 	2024
Graduate Dean's Scholars Award , UCLA scholarship.	2024
Awardee , NTSE Scholar (India).	2014-2020